

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**2<sup>nd</sup> Semester of B.Pharm. Examination**  
**University Theory Examination May 2017**  
**MA122 Advanced Mathematics**

**Date: 17.05.2017, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80**

**Instructions:**

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.
6. Use of scientific calculator is allowed.

**SECTION - I**

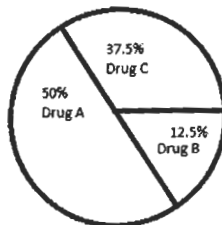
**Q 1** Attempt all questions. Each question is of one mark. **20**

1. Fifth derivative of  $y = e^{2x}$  is  
 [A]  $y_5 = 2^5 e^{2x}$ .  
 [B]  $y_5 = 2^{-5} e^{2x}$ .  
 [C]  $y_5 = e^{2x+5}$ .  
 [D]  $y_5 = 2^{-5} e^{2x+5}$ .
2. 15<sup>th</sup> derivative of  $y = (x + 5)^4$  is  
 [A] 1.  
 [B] 3.  
 [C] 0.  
 [D] 4.
3. Which of the following is true?  
 [A] Every Taylor series is Macluarin series.  
 [B] Every Macluarin series is Taylor series.  
 [C] Macluarin series expansion around 1 is Taylor series.  
 [D] Macluarin series expansion around 2 is Taylor series.
4. The order and degree of an ordinary differential equation  $\left(\frac{d^2 y}{dx^2}\right)^{\frac{1}{2}} - y = 0$  are  
 [A] order 1 and degree 1.  
 [B] order 2 and degree 1.  
 [C] order 1 and degree 2.  
 [D] order 2 and degree 2.

5. The first order ordinary differential equation  $M(x, y)dx + N(x, y)dy = 0$  is exact if and only if
- [A]  $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$ .
- [B]  $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$ .
- [C]  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ .
- [D]  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$ .
6. The general form of linear differential equation is
- [A]  $\frac{dy}{dx} + p(y)y = q(x, y)$ .
- [B]  $\frac{dy}{dx} + p(x, y)y = q(y)$ .
- [C]  $\frac{dy}{dx} + p(x)y^n = q(x), n > 1$ .
- [D]  $\frac{dy}{dx} + p(x)y = q(x)$ .
7. Order of homogeneous function  $f(x, y) = x^4 + 2x^3y + 5x^2y^2 + y^4$  is
- [A] 1.
- [B] 2.
- [C] 3.
- [D] 4.
8. Which of the following is a particular integral of  $y'' + 4y' + 4y = 0$ ?
- [A] 0.
- [B] 1.
- [C]  $xe^{-2x}$ .
- [D]  $e^{-x}$ .
9. If roots of auxiliary equation are 1 and 2, then corresponding component (basis) solutions are
- [A]  $e^x$  &  $e^{2x}$ .
- [B]  $e^{-x}$  &  $e^{-2x}$ .
- [C]  $e^x$  &  $e^{-2x}$ .
- [D]  $e^{-2x}$  &  $e^{3x}$ .
10. Which of the following is true for the rate of change in number of bacteria(x) with respect to time(t) is inverse propositional to time?
- [A]  $\frac{dx}{dt} = mt$ .
- [B]  $t \frac{dx}{dt} = m$ .
- [C]  $\frac{dx}{dt} = mt^2$ .
- [D]  $\frac{dx}{dt} = mt + 1$ .

11. The rate at which bacteria multiply is proportional to the current number of bacteria. if  $x$  be the number of bacteria present at time  $t$ , then
- [A]  $\frac{dx}{dt} = mx$ .
- [B]  $\frac{dx}{dt} = mx^2$ .
- [C]  $\frac{dx}{dt} = mx + kt$ .
- [D]  $\frac{dx}{dt} = e^{mx}$ .
12. If  $T$  be the temperature of the body at time  $t$  and  $t_s$  is the temperature of surrounding, then Newton's law of cooling is given by
- [A]  $\frac{dT}{dt} = k(T - t_s)$ .
- [B]  $\frac{dT}{dt} = kT + t_s$ .
- [C]  $\frac{dT}{dt} = kT$ .
- [D]  $\frac{dT}{dt} = k(t_s - T)$ .
13. Equation of a straight line passing through  $(1, 0)$  and  $(0, 1)$  is
- [A]  $x + y = 1$ .
- [B]  $x - y = 1$ .
- [C]  $x + y = 0$ .
- [D]  $x - y = 0$ .
14. Equation of a straight line parallel to  $y = 4x + 3$  is
- [A]  $y + 4x = 3$ .
- [B]  $y = 2x - 5$ .
- [C]  $y = 2x + 3$ .
- [D]  $y = 4x + 5$ .
15. Two lines with slope  $m_1$  &  $m_2$  are said to be perpendicular if
- [A]  $m_1 m_2 = 1$ .
- [B]  $m_1 - m_2 = 0$ .
- [C]  $m_1 + m_2 = 1$ .
- [D]  $m_1 m_2 = -1$ .
16. Distance between two points  $(1, 2)$  and  $(1, 4)$  is
- [A] 0.
- [B] 2.
- [C] -2.
- [D] 3.
17. Mean of data set 1, 6, 8, 7, 10, 10, 12, 12 and 15 is
- [A] 8.
- [B] 9.
- [C] 10.
- [D] 11.

18. Data set 2, 6, 7,  $a$ , 12, 14 and 15 is arranged in increasing order. If Median of a following data set is 8, then  $a$  is  
 [A] 10.  
 [B] 11.  
 [C] 8.  
 [D] 9.
19. Following graph shows production of different types of drug by a company in a month. If total production of drugs by a company is 12 kg in one month, then production of drug B in a month is



- [A] 1.5 kg.  
 [B] 6 kg.  
 [C] 4.5kg.  
 [D] 10 kg.
20. If mean of data set 5, 8, 10, 12,  $a$  is 10, then value of  $a$  is  
 [A] 13.  
 [B] 14.  
 [C] 15.  
 [D] 16.

### SECTION – II

**Q 2** Attempt any TWO of the following

- A** Bacteria multiply at the rate proportional to the number present. If the original number  $N$  doubles in three hours, then find required time so that the number of bacteria will be  $4N$ . **05**
- B** Find mean, median and relative frequency distribution(%) of ungrouped data set 1, 5, 7, 8, 2, 3, 7, 2, 8 and 7. **05**
- C** Derive equation of the distance between two points and using it find distance between (3, 4) and (-1, -2). **05**

**Q 3** Attempt any TWO of the following

- A** Verify Cauchy's mean value theorem for  $f(x) = x^3$  and  $g(x) = 2 - x$ ,  $x \in [0, 2]$ . **05**
- B** Identify the type of an ordinary differential equation  $\frac{dy}{dx} + 4\frac{y}{x} = x^3$  and find particular solution with help of initial condition  $y(1) = 2$ . **05**
- C** Find general solution of  $y'' + 2y' + 4y = 0$ . **05**

**SECTION – III****Q 4** Attempt any **FOUR** of the following

- A** Differential equation  $\frac{ds}{dt} + 4s = e^t$  represents a relation between sugar(s) level 05  
in blood and time(t) of a person. Solve the differential equation for sugar(s)  
level in term of t.
- B** The population of a community is known to increase at a rate proportional to 05  
the number of people present at a time t. If the population has doubled in 6  
years, how long it will take to triple?
- C** Obtain a perpendicular line to  $y = x$  which passes through (2, -2) and a parallel 05  
line to  $y = x + 1$  which passes through (1, 1). Find intersect point of these  
obtained lines.
- D** A rectangle having vertices A(0,0), B(2,0), C(2, 3) and D(0, 3) is given. Find 05  
length, height and area of rectangle.
- E** Find Mean for following ungrouped data. 05  
25 27 29 18 22 26 31 27 32 34 35 36 39 40 41 41 42 43  
Also organize the above data into groups with 5 intervals starting from  
smallest value and using it evaluate relative frequency(%) of each class.
- F** (I) If the rate of change of population is propotional to time and at 1 hour 03  
rate is given by 100 person per hour, then find number of person at  $t = 3$   
hour where at intial stage number of person is zero.  
(II) Describe limitation of mean, median and mode. 02

**Q 5** Attempt any **FOUR** of the following

- A** Find Maclaurin series of  $\sin x$  and  $\cos x$ . 05
- B** Find Taylor series of function  $f(x) = x^3 + 6x + 10$  around  $x = 2$ . Also show 05  
that Maclaurin series is a particular case of Taylor series.
- C** Solve differential equation  $\frac{dy}{dx} = \frac{x^3 + 2xy^2}{x^2 y}$ . 05
- D** Show that  $2xydx + (x^2 + y^2 + 2)dy = 0$  is an exact differential equation and 05  
obtain general solution of it.
- E** Find solution of  $y'' - y = e^x$ . 05
- F** Solve  $y'' + 5y' + 6y = e^{3x}$ . 05